

EXERCISE - 5.1

Multiple choice questions

1. Which of the following is true ?

(1) $\frac{x^2+2}{\sqrt{x^2+1}} \geq 5$ (2) $\frac{x^2+2}{\sqrt{x^2+1}} \leq 2$ (3) $\frac{x^2+2}{\sqrt{x^2+1}} \geq 2$ (4) $\frac{x^2+2}{\sqrt{x^2+1}} \leq 5$

2. If $0 < a < 1$, then the value of the expression

$$\left\{ \frac{\sqrt{1+a}}{\sqrt{1+a}-\sqrt{1-a}} + \frac{1-a}{\sqrt{1-a^2}-1+a} \times \sqrt{\frac{1}{a^2}-1} - \frac{1}{a} \right\}$$

lies between

(1) -1 and +1 (2) -2 and +2 (3) -3 and +3 (4) -4 and +4

3. Which is bigger 17^{14} or 31^{11} .

(1) $17^{14} > 31^{11}$ (2) $17^{14} < 31^{11}$ (3) $17^{14} = 31^{11}$ (4) None of these

4. Without using tables, find that $\frac{1}{\log_2 \pi} + \frac{1}{\log_5 \pi}$ is always greater than

(1) 0 (2) 1 (3) $3/2$ (4) 2

5. Let a, b, c be real numbers with $0 < a < 1, 0 < b < 1, 0 < c < 1$ and $a + b + c = 2$ then $\frac{a}{1-a} \cdot \frac{b}{1-b} \cdot \frac{c}{1-c}$ is always greater than or equal to

(1) 2 (2) 4 (3) 6 (4) 8

6. If a, b, c are positive and $a + b + c = 1$. Then $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ is greater than or equal to

(1) 6 (2) 9 (3) 12 (4) 8

7. If a, b, c, d are four non-negative real numbers and $a + b + c + d = 1$ then $ab + bc + cd$ is less than equal to

(1) 1 (2) 2 (3) $3/2$ (4) $1/4$

8. For any three positive real numbers a, b and c , find the sign of $a^2 + b^2 + c^2 - ab - bc - ca$

(1) is positive or equal to zero (2) is negative or equal to zero
(3) is neither positive or negative (4) can not be found

9. For any real numbers x, y, z find minimum value of $x^4 + y^4 + z^4 - x^2y^2 - x^2z^2 - y^2z^2$

(1) 0 (2) 1 (3) 2 (4) 3

10. For positive a, b, c , find minimum value of $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}$

(1) $\frac{1}{2}$ (2) 1 (3) $\frac{3}{2}$ (4) 2

11. If a, b, c, d are positive real numbers find minimum value of $(a+b+c+d) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right)$

(1) 0 (2) 4 (3) 8 (4) 16

12. If x, y, z are positive unequal reals $x < y < z$, then which of the following is true ?

$$(1) \frac{x^2}{z} > \frac{x^2 + y^2 + z^2}{x + y + z} > \frac{z^2}{x}$$

$$(2) \frac{x^2}{z} > \frac{x^2 + y^2 + z^2}{x + y + z} \text{ OR } \frac{x^2 + y^2 + z^2}{x + y + z} < \frac{z^2}{x}$$

$$(3) \frac{x^2}{z} < \frac{x^2 + y^2 + z^2}{x + y + z} > \frac{z^2}{x}$$

$$(4) \frac{x^2}{z} < \frac{x^2 + y^2 + z^2}{x + y + z} < \frac{z^2}{x}$$

13. n is a natural number greater than 1, and

$$A = \frac{\sqrt{n+1}}{n} + \frac{\sqrt{n+4}}{n+3} + \frac{\sqrt{n+7}}{n+6} + \frac{\sqrt{n+10}}{n+9} + \frac{\sqrt{n+13}}{n+12}$$

$$B = \frac{1}{\sqrt{n-1}} + \frac{1}{\sqrt{n+2}} + \frac{1}{\sqrt{n+5}} + \frac{1}{\sqrt{n+8}} + \frac{1}{\sqrt{n+11}}, \text{ then}$$

$$(1) A = B$$

$$(2) A = 2B$$

$$(3) A < B$$

$$(4) A > B$$

14. The minimum value of the terms of the sequence

$$\sqrt{\frac{7}{6}} + \sqrt{\frac{96}{7}}, \sqrt{\frac{8}{6}} + \sqrt{\frac{96}{8}}, \sqrt{\frac{9}{6}} + \sqrt{\frac{96}{9}}, \dots, \sqrt{\frac{95}{6}} + \sqrt{\frac{96}{95}} \text{ is}$$

$$(1) 6$$

$$(2) 7$$

$$(3) 8$$

$$(4) 4$$

15. a and b are the roots of the quadratic equation $x^2 + \lambda x - \frac{1}{2\lambda^2} = 0$ where x is the unknown and λ is a real parameter. The minimum value of $a^4 + b^4$ is

$$(1) 2\sqrt{2}$$

$$(2) \frac{2}{1+\sqrt{2}}$$

$$(3) \sqrt{2}$$

$$(4) 2 + \sqrt{2}$$

16. For positive unequal a and b , then which of the following is true ?

$$(1) 3ab^2 > a^3 + 2b^3$$

$$(2) 3ab^2 < a^3 + 2b^3$$

$$(3) 3ab^2 = a^3 + 2b^3$$

$$(4) \text{None of these}$$

17. If a and b are real positive distinct numbers. Then which one is true ?

$$(1) a^3b + ab^3 > a^4 + b^4$$

$$(2) a^3b + ab^3 = a^4 + b^4$$

$$(3) a^3b + ab^3 < a^4 + b^4$$

$$(4) \text{None of these}$$

18. If a and b are positive real numbers such that $ab = 1$, then the least value of the expression $(1+a)(1+b)$ is

$$(1) 2$$

$$(2) 3$$

$$(3) 4$$

$$(4) 5$$

19. If a, b, c are the sides of a triangle, then

$$\frac{a}{b+c-a} + \frac{b}{c+a-b} + \frac{c}{a+b-c}$$

$$(1) \leq 3$$

$$(2) \geq 3$$

$$(3) \geq 2$$

$$(4) \leq 2$$

20. If $a^2 + b^2 + c^2 = 1$, then for $ab + bc + ca$, which one is true ?

$$(1) -\frac{1}{2} \leq ab + bc + ca \leq 1$$

$$(2) \frac{1}{2} \leq ab + bc + ca \leq 1$$

$$(3) -2 \leq ab + bc + ca \leq 1$$

$$(4) -2 \leq ab + bc + ca \leq -1$$

21. The function $f(n)$ is defined for all positive integer n and take on non-negative integer values such that $f(2) = 0$, $f(3) > 0$ and $f(9999) = 3333$. Also, for all m, n ,

$$f(m+n) - f(m) - f(n) = 0 \quad \text{or} \quad 1.$$

Determine sum of all the digits of $f(2005)$.

22. Let x and y be positive integers such that $\frac{100}{151} < \frac{y}{x} < \frac{200}{251}$. What is the minimum value of x ?
23. Let $f(x) = ax^2 + bx + c$, where a, b and c are integers. Suppose that $f(1) = 0$, $50 < f(7) < 60$, $70 < f(8) < 80$, and $5000k < f(100) < 5000(k+1)$ for some integer k . What is $3k^2$?
24. Given that x and y are positive integers such that $56 \leq x + y \leq 59$ and $0.9 < \frac{x}{y} < 0.91$, find the value of

$$\left(y + \frac{x}{4}\right)$$

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	1	4	4	2	4	1	1	3	4	4	3	4	4
Que.	16	17	18	19	20										
Ans.	2	3	3	2	1										

21. 20

22. 03

23. 27

24. 38

Important Notes

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